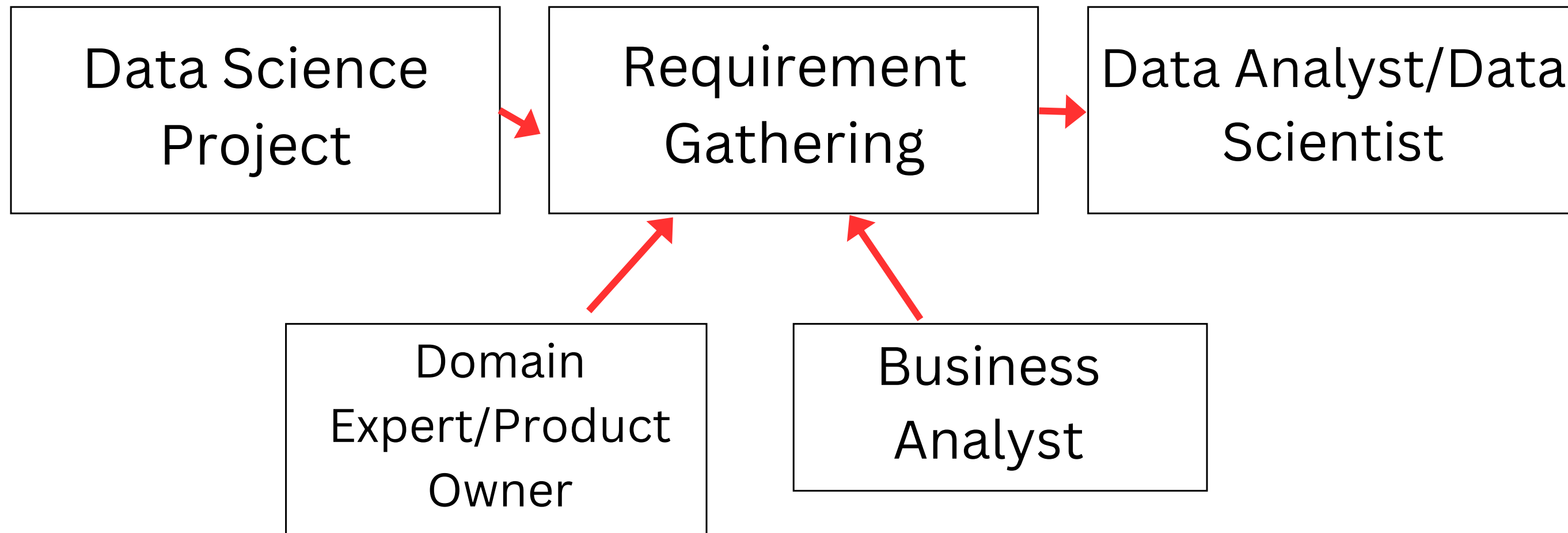


Data Engineering using Pandas

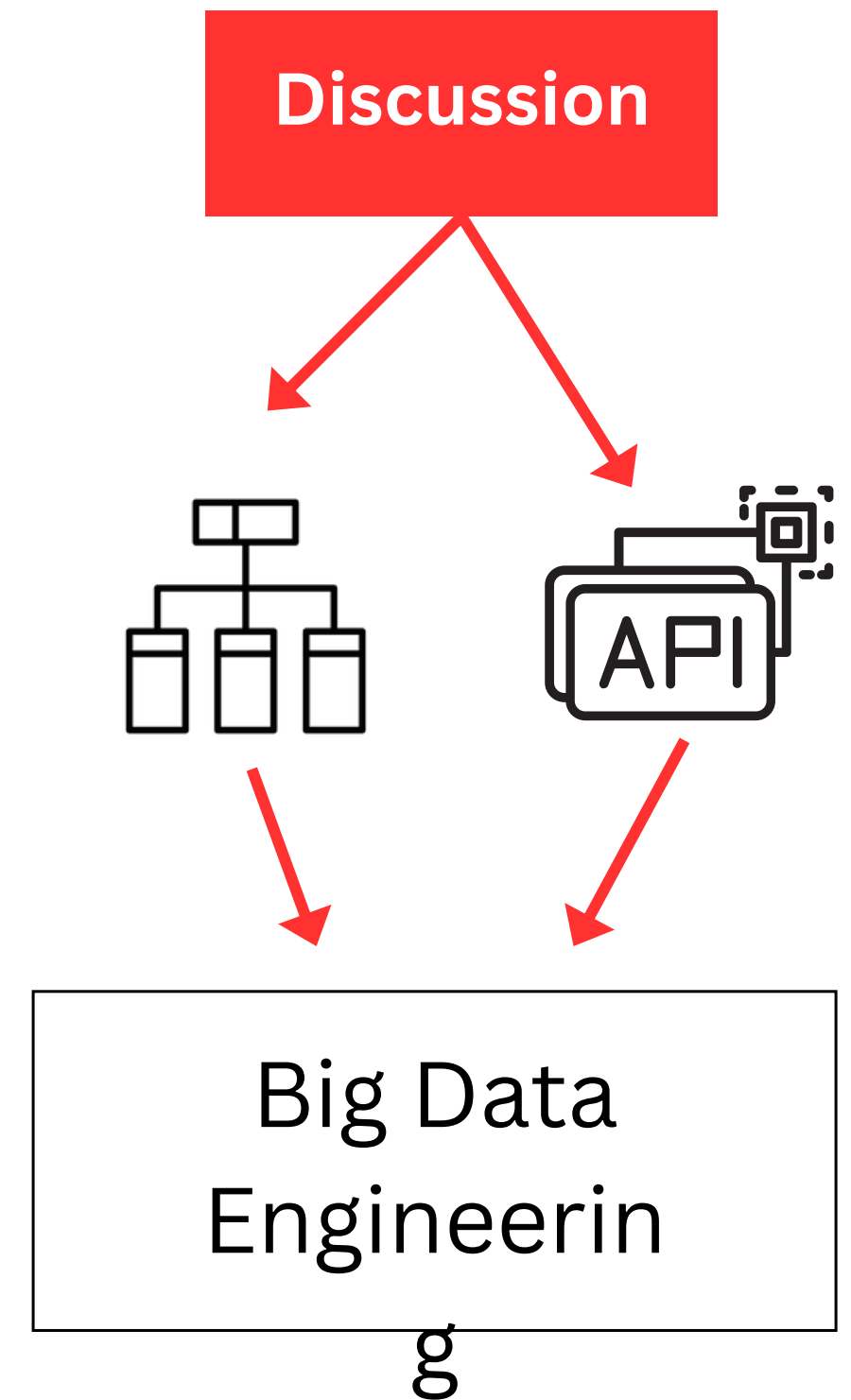
Week 3

MLOPS Stacks



Machine Learning Operations

2



ML Process

Data Collection



Features
Engineering



Features
Selection



Model Creation &
HyperParameters
Tunning



Model
Deployment



Model Monitoring
& Training

- **EDA**
- **Missing Values**
- **Data Duplicates**
- **Outliers**
- **Normalization**
- **Encoding**

Machine Learning Operations

- **Correlation**
- **Forward Selection**
- **PCA**

3

- **Linear/Logistic Regression**
- **DecisionTrees**
- **RandomForest**
- **KNN**
- **ANN/CNN/LSTM**

Exploratory Data Analysis

```
df.describe()
```

	client_id	loan_amount	repaid	loan_id	rate
count	443.000000	443.000000	443.000000	443.000000	443.000000
mean	38911.060948	7982.311512	0.534989	11017.101580	3.217156
std	7768.681063	4172.891992	0.499338	581.826222	2.397168
min	25707.000000	559.000000	0.000000	10009.000000	0.010000
25%	32885.000000	4232.500000	0.000000	10507.500000	1.220000
50%	39505.000000	8320.000000	1.000000	11033.000000	2.780000
75%	46109.000000	11739.000000	1.000000	11526.000000	4.750000
max	49624.000000	14971.000000	1.000000	11991.000000	12.620000

Machine Learning Operations

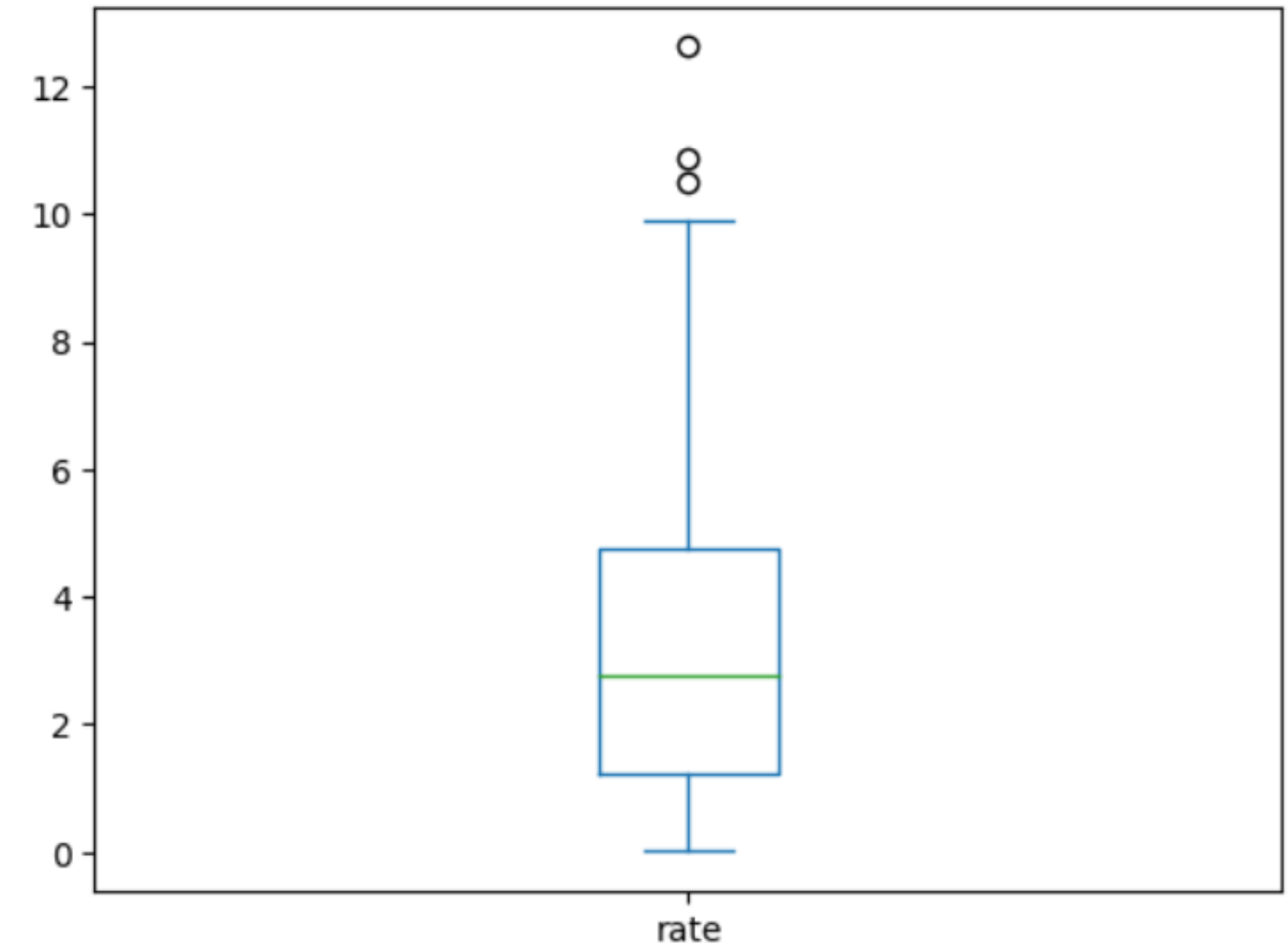
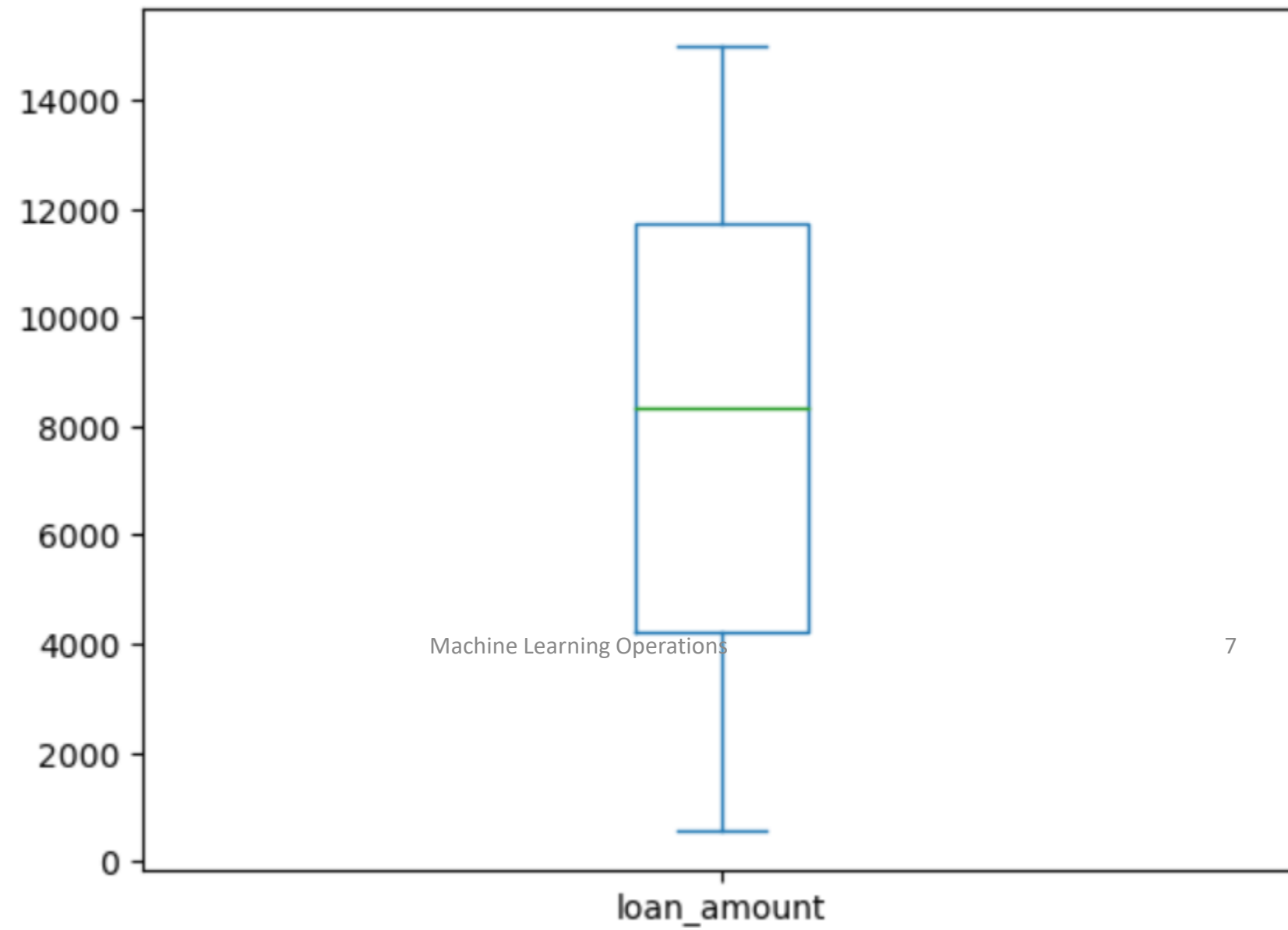
Exploratory Data Analysis

- Let's an array is: [3, 7, 8, 5, 12, 14, 21, 13, 18, 20]
- **Sorted:** [3, 5, 7, 8, 12, 13, 14, 18, 20, 21]
- **count** = 10
- **mean** =
 $(3+7+8+5+12+14+21+13+18+20)/10=12.1$
- **Std**= 6.2973
- **min** = 3
- **25% (Q1)** — $h=(n-1)p+1=(9)(0.25)+1=3.25$ Machine Learning Operations $h=(n-1)p+1=(9)(0.25)+1=3.25 \rightarrow$
between 3rd (7) & 4th (8): $7+0.25(8-7)=7.25$
- **50% (median)** $h=9(0.5)+1=5.5$ $h=9(0.5)+1=5.5 \rightarrow$ avg of 5th (12) & 6th (13): 12.5
- **75% (Q3)** $h=9(0.75)+1=7.75$ $h=9(0.75)+1=7.75 \rightarrow$ between 7th (14) & 8th (18): $14+0.75(18-14)=17.0$
- **max** = 21

Outliers

- Outliers are unusual values in your dataset that are very different from most other values. They can be much larger or much smaller than the majority of data points.
- An outlier is an observation that lies an abnormal distance from other values in a random sample from a population.
- Methods to detect outliers
 - a) Visual Methods** (Boxplot, Bar plots, Violin plots, Histograms)
 - b) Statistics Methods** (Z-Score, IQR)
 - c) Model Based Methods** (Kmeans, Isolation Forests)

BoxPlots



BoxPlot

Z-Score Methods

Z – Score formula is: $Z = (X_i - \text{mean}) / \text{std}$

X_i = data point

Mean= mean of the data

Std: Standard Deviation of the data

A **Z-score of 0** shows that the data point is exactly at the mean.

A **positive Z-score** shows that the data point is above the mean.

A **negative Z-score** shows that the data point is below the mean

Z-score greater than **3** or less than **-3** are considered outliers

Z-Score

Example

- Lets an array is: [10, 12, 12, 13, 12, 14, 13, 100]
- Calculating mean and count:
 $n = 8$
 $\text{sum} = 10 + 12 + 12 + 13 + 12 + 14 + 13 + 100 = 186$
 $\text{mean} = \text{sum} / n = 186 / 8 = 23.25$
- Calculate deviations from mean $\Rightarrow (x_i - \text{mean})$ & $(x_i - \text{mean}) * (x_i - \text{mean})$
- Calculate z-score of each data sample.
- If for any data sample z-score is greater than 3 or less than -3 that sample is called as **Outliers**

Z-Score

Code

```
import numpy as np
arr = np.array([10,12,12,13,12,14,13,100])
mean = arr.mean()
std = arr.std(ddof=1)
z = (arr - mean) / std
outliers_z = arr[np.abs(z) > 3]
# modified z-score (robust)
median = np.median(arr)
mad = np.median(np.abs(arr - median))
modified_z = 0.6745 * (arr - median) / mad
outliers_modz = arr[np.abs(modified_z) > 3.5]
```

Calculating Mean

Calculate Standard Deviation

Z Score to each element

Inter Quartile Range (IQR)

- Formula for IQR (Interquartile Range) $IQR=Q3-Q1$

- Where: **Q1** = 25th percentile (first quartile)

Q3 = 75th percentile (third quartile)

using IQR: **Lower Bound**= $Q1-1.5\times IQR$

Upper Bound= $Q3+1.5\times IQR$

Any value **less than Lower Bound** or **greater than Upper Bound** is considered an outlier.

Inter Quartile Range (IQR)

- Let's array is: [5, 5, 5, 6, 6, 7, 7, 8, 8, 100]
- **Q1** (25th percentile): value between 2nd and 3rd index
(since 25% of 10 = 2.5) $\rightarrow (5 + 6) / 2 = 5.5$
- **Q2** (median, 50%): value between 5th and 6th index
(since 50% of 10 = 5.0) $\rightarrow (6 + 7) / 2 = 6.5$
- **Q3** (75th percentile): value between 7th and 8th index
(since 75% of 10 = 7.5) $\rightarrow (7 + 8) / 2 = 7.5$
- **IQR = Q3 - Q1 = 7.5 - 5.5 = 2.0**
- **Lower bound = Q1 - 1.5 × IQR = 5.5 - (1.5 × 2.0) = 5.5 - 3.0 = 2.5**
- **Upper bound = Q3 + 1.5 × IQR = 7.5 + (1.5 × 2.0) = 7.5 + 3.0 = 10.5**

Inter Quartile Range (IQR)

```
import numpy as np
data = np.array([10, 12, 12, 13, 12, 14, 13, 120, 15, 11])
Q1 = np.percentile(data, 25)           # Calculate Q1 (25th percentile)
Q3 = np.percentile(data, 75)           # Q3 (75th percentile)
IQR = Q3 - Q1                          # Step 2: Calculate IQR
lower_bound = Q1 - 1.5 * IQR            # Step 3: Define bounds
upper_bound = Q3 + 1.5 * IQR
outliers = data[(data < lower_bound) | (data > upper_bound)] # Step 4: Detect outliers
```


MLOps Stacks



Flask



PostgreSQL

ASTRONOMER



Amazon SageMaker